

Mathematics Reference: Profit and Loss

Educational Series

December 29, 2025

1 Basic Definitions

- **Cost Price (C.P.):** The price at which an article is purchased.
- **Selling Price (S.P.):** The price at which an article is sold.
- **Marked Price (M.P.):** The price listed on the label of an article (also known as List Price).

2 Fundamental Formulas

2.1 Profit and Loss

- **Profit (Gain):** Occurs when $S.P. > C.P.$

$$\text{Profit} = S.P. - C.P.$$

- **Loss:** Occurs when $C.P. > S.P.$

$$\text{Loss} = C.P. - S.P.$$

2.2 Percentage Formulas

Profit and Loss percentages are **always** calculated on the Cost Price (C.P.) unless otherwise stated.

$$1. \text{ Profit \%} = \left(\frac{\text{Profit}}{C.P.} \times 100 \right) \%$$

$$2. \text{ Loss \%} = \left(\frac{\text{Loss}}{C.P.} \times 100 \right) \%$$

3 Finding S.P. and C.P.

3.1 When Profit % is given:

$$S.P. = \left[\frac{100 + \text{Profit \%}}{100} \right] \times C.P.$$

$$C.P. = \left[\frac{100}{100 + \text{Profit \%}} \right] \times S.P.$$

3.2 When Loss % is given:

$$S.P. = \left[\frac{100 - \text{Loss \%}}{100} \right] \times C.P.$$

$$C.P. = \left[\frac{100}{100 - \text{Loss \%}} \right] \times S.P.$$

4 Discount and Marked Price

Discounts are **always** calculated on the Marked Price (M.P.).

- $S.P. = M.P. - \text{Discount}$
- $\text{Discount \%} = \left(\frac{\text{Discount}}{M.P.} \times 100 \right) \%$
- $S.P. = M.P. \times \left(\frac{100 - \text{Discount \%}}{100} \right)$

5 Successive Discounts

If an article is sold at successive discounts of $x\%$ and $y\%$, the effective single discount is:

$$\text{Equivalent Discount} = \left(x + y - \frac{xy}{100} \right) \%$$

6 Solved Example

Problem: A man buys a watch for \$800 and sells it for \$1000. Find his profit percentage.

Solution:

- $C.P. = 800, S.P. = 1000$
- $\text{Profit} = 1000 - 800 = 200$
- $\text{Profit \%} = \left(\frac{200}{800} \times 100 \right) \% = 25\%$